

Chapter 1

Static Force Analysis

A thorough understanding of the forces that act on the members of a machine or mechanism is vital for designing them effectively.

Static force analysis deals with the effect of externally applied forces on machine members. The mass and acceleration of the mechanism is not considered. This type of analysis is quite relevant for low/medium speed engines.

1.1

Conditions for equilibrium

The essential conditions that a body or machine member in static equilibrium must satisfy are

(1) **Force law of equilibrium**

$$\Sigma H = 0 \quad \text{i.e. sum of horizontal forces} = 0$$

$$\Sigma V = 0 \quad \text{i.e. sum of vertical forces} = 0$$

(2) **Moment law of equilibrium**

$$\Sigma M = 0 \quad \text{i.e. sum of moments} = 0$$

1.2

Static equilibrium of different types of members

Depending on the kind of member considered, the conditions for static equilibrium can be set forth as follows:

Two-force members For a two-force member to be in equilibrium

- (1) The forces should be equal and opposite.
- (2) They should be collinear i.e. they must act along the same line.

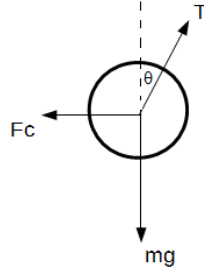


FIGURE 1. Equilibrium of a two-force member

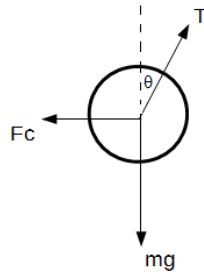


FIGURE 2. Equilibrium of a three-force member

Three-force members A three-force member is in static equilibrium if

- (1) The forces are all concurrent - i.e. the lines of action of the forces must meet at a single point.
- (2) The vector diagram of the forces form a closed triangle - i.e. there is no resultant force.

For equilibrium, both these conditions must be satisfied by the system of forces. They will also obey Lami's theorem.

Four-force members A four-force member is in equilibrium if

- (1) There is no resultant - i.e. the forces form a closed polygon
- (2) The line of action of any two forces and the resultant of the other two intersect at a point - i.e. they are concurrent.

Member with two forces and applied torque In this case, the forces must be equal, opposite and parallel to each other so that they form a couple. This couple should be equal in magnitude and opposite in sense to the applied torque.



i.e. For equilibrium,

Applied torque, $T = \text{Resultant couple, } C = F \times h$

1.3

Principle of Superposition of Forces

This principle states that the net effect of the action of a number of forces on a linear system is the sum of the effects of the individual forces taken one at a time.

1.4

Static force analysis using principle of superposition

The main objective of static force analysis is to determine the *constraint forces or torques* required to keep a linkage in equilibrium under the action of an applied load.

If there is only a single force acting on the system, the entire system can be considered as one and the conditions for equilibrium can be applied.

If multiple forces act, the action of each individual force is considered separately and the effects superimposed.

Naming of forces Force exerted by member A on member B can be written as F_{AB} .

Also,

$$F_{AB} = -F_{BA}$$

Free Body Diagram The free body diagram of a member is a diagram that shows the magnitude and direction of all external forces acting on that member alone.

1.5

Static force analysis without considering friction

Static force analysis can be done using graphical or vector method. Both methods are illustrated in the example given below

EXAMPLE 1. *Determine the forces and torques required to hold the four bar linkage given below in equilibrium is a force $F = 240 \text{ N}$ acts on it as shown.*

The dimensions are $AB = 120 \text{ mm}$ $BC = 360 \text{ mm}$ $CD = 240 \text{ mm}$
 $AD = 160 \text{ mm}$ $DE = 100 \text{ mm}$



SOLUTION. **GRAPHICAL METHOD**

Step 1 : Draw the given linkage to a suitable scale



Step 2: Consider each link in turn

Link 2

At joint B, F_{32} acts along the direction BC and F_{12} acts at C. Since 2 is in equilibrium under two forces F_{32} and F_{12} and a torque T , from conditions for equilibrium, F_{32} and F_{12} must be equal and parallel so that

$$C = F_{32} \times h = F_{12} \times h = T \quad (1.1)$$

Thus F_{12} can be marked in this direction.

Link 3

BC is a two-force member in equilibrium. So, there will be two equal and opposite forces acting along BC.



FIGURE 3. Link 2



FIGURE 4. Link 3



FIGURE 5. Link 4

Link 4

Link 4 or CD is a three-force member in equilibrium under the action of three forces F_{34} , F_{14} and F of which F is completely known.

The lines of action of F_{34} along BC and F are known. From conditions for equilibrium the lines of action of all three forces must meet. Hence the direction of F_{14} is obtained. On completing the force polygon, the magnitude of these forces are also determined.

Steps to draw force polygon

FIGURE 6. Force polygon

- (1) Draw vector F having magnitude 240 N to suitable scale along the given direction as pq
- (2) Draw lines parallel to F_{34} and F_{14} to meet at point r
- (3) The lengths pr and qr will be proportional to the magnitudes of F_{34} and F_{14}

From force polygon, $F_{34}=60$ N, $F_{14} = 192$ N

Also, $F_{34}=F_{43}=F_{23}=F_{32}$

From diagram,

$h = 1.5 \text{ cm, equivalent to } 120 \text{ mm}$

$$C = F_{32} \times h = 60 \times 120 = 7200 \text{ N-mm} = 7.2 \text{ N-m counter clockwise}$$

$\therefore$ The torque T required to keep the linkage in equilibrium = **7.2 N-m clockwise**

VECTOR METHOD

To use this method, the angles made by each link must be known.

From fig.(i), the angles made by each link with horizontal - link 1 - is measured.

- Link 2, $\theta_2 = 135^\circ$
- Link 3, $\theta_3 = 22.5^\circ$
- Link 4, $\theta_4 = 68.5^\circ$

Note: For any link PQ as shown below, the position vector can be written as



Position of Q with respect to P

$$\vec{r}_{qp} = l \left(\cos \theta \hat{i} + \sin \theta \hat{j} \right)$$

Similarly for the given linkage

$$\vec{r}_{ba} = 120 \left(\cos 135^\circ \hat{i} + \sin 135^\circ \hat{j} \right)$$

$$= -84.85\hat{i} + 84.85\hat{j}$$

$$\vec{r}_{cb} = 360 \left(\cos 22.5^\circ \hat{i} + \sin 22.5^\circ \hat{j} \right)$$

$$= 332.6\hat{i} + 137.8\hat{j}$$

$$\vec{r}_{cd} = 240 \left(\cos 68.5^\circ \hat{i} + \sin 68.5^\circ \hat{j} \right)$$

$$= 87.9\hat{i} + 223.3\hat{j}$$

$$\vec{r}_{ed} = 100 \left(\cos 68.5^\circ \hat{i} + \sin 68.5^\circ \hat{j} \right)$$

$$= 36.65\hat{i} + 93.04\hat{j}$$

The force vectors on the link 4 are

$$\vec{F}_{34} = F_{34} (\cos 22.5^\circ \hat{i} + \sin 22.5^\circ \hat{j}) = F_{34} (0.924\hat{i} + 0.383\hat{j})$$

The angles are measured at point A. So when F is extended and shifted, θ becomes $180 + 40 = 220^\circ$.



Taking moments about point D for link 4, for equilibrium,

$$\Sigma \vec{M}_D = 0$$

$$\Sigma \vec{M}_D = \vec{r}_{cd} \times \vec{F}_{34} + \vec{r}_{ed} \times \vec{F} = 0$$

The cross product can be obtained as

$$\begin{aligned} \vec{r}_{cd} \times \vec{F}_{34} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 87.9 & 223.3 & 0 \\ 0.924F_{34} & 0.383F_{34} & 0 \end{vmatrix} \\ &= \hat{i} (223.3 \times 0 - 0.383F_{34} \times 0) \\ &\quad - \hat{j} (87.9 \times 0 - 0.924F_{34} \times 0) \\ &\quad + \hat{k} (87.9 \times 0.383 - 223.3 \times 0.924) F_{34} \\ &= -172.66F_{34}\hat{k} \end{aligned}$$

Similarly,

$$\begin{aligned} \vec{r}_{ed} \times \vec{F} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 36.65 & 93.04 & 0 \\ -183.8 & -154.27 & 0 \end{vmatrix} \\ &= \hat{i} (93.04 \times 0 - -154.27 \times 0) \\ &\quad - \hat{j} (36.65 \times 0 - -183.8 \times 0) \\ &\quad + \hat{k} (36.65 \times -154.27 - 93.04 \times -183.8) \\ &= 11446.8\hat{k} \\ \therefore \Sigma \vec{M}_D &= 11446.8\hat{k} - 172.66F_{34}\hat{k} = 0 \\ F_{34} &= 66.3 \text{ N} \end{aligned}$$

Hence the vector

$$\begin{aligned}\vec{F}_{34} &= 66.3 \left(0.924\hat{i} + 0.383\hat{j} \right) \\ &= \left(61.3\hat{i} + 25.4\hat{j} \right)\end{aligned}$$

Link 4 has three forces acting on it - $\vec{F}_{34}$, $\vec{F}_{14}$ and $\vec{F}$

For equilibrium,

$$\begin{aligned}\Sigma F &= 0 = \vec{F}_{34} + \vec{F}_{14} + \vec{F} = 0 \\ \Rightarrow -183.8\hat{i} - 154.27\hat{j} + 61.3\hat{i} + 25.4\hat{j} + \vec{F}_{14} &= 0 \\ \therefore \vec{F}_{14} &= \left(122.5\hat{i} + 128.9\hat{j} \right) N \\ \vec{F}_{34} = -\vec{F}_{43} = \vec{F}_{23} = -\vec{F}_{32} = \vec{F}_{12} &= \left(61.3\hat{i} + 25.4\hat{j} \right) N \\ \therefore \vec{F}_{32} = -\vec{F}_{23} &= -\left(61.3\hat{i} + 25.4\hat{j} \right) N\end{aligned}$$

Now, only torque T is unknown.

Consider link 2

Taking moments about A ,

$$\Sigma M_A = 0 \Rightarrow \vec{r}_{ba} \times \vec{T} = 0$$

$\therefore \vec{T} = 7.35 \text{ N-m perpendicular to } XY \text{ plane and clockwise in direction}$

1.6

Solved Problems - Set-I

PROBLEM 1.1. A slider crank mechanism has a crank of length 150 mm and connecting rod of length 675 mm. The piston is acted upon by a force $P = 3 \text{ kN}$ as shown. Determine the torque T to be provided to the crank to keep the mechanism in static equilibrium when the crank makes an angle of 120° with the line of travel of the piston.



SOLUTION. GRAPHICAL METHOD

Consider the equilibrium of each link in the mechanism

Link 1 is fixed.

Link 2 is under the action of two forces and a torque F_{32} , F_{12} and T



From the conditions for equilibrium, $F_{12} = -F_{32}$ and $T = C = F_{32} \times h$

Link 3 is a two-force member under the action of F_{23} and F_{43}

Link 4 is a three-force member under the action of P , F_{34} and F_{14}

Only P is fully known. F_{14} will be perpendicular to the line of slider travel, and F_{34} acts along AB as shown in fig. iii.

The magnitudes are found from the force polygon. From the force polygon in fig.iv,

$$F_{14} = 0.6 \text{ kN and } F_{34} = 3.06 \text{ kN}$$

$$\therefore F_{43} = -3.06 \text{ kN}$$

$$F_{23} = -F_{43} = 3.06 \text{ kN}$$

$$F_{32} = -F_{23} = -3.06 \text{ kN}$$

$$F_{12} = -F_{32} = 3.06 \text{ kN}$$

From fig. i, $h = 115 \text{ mm}$

Also,

The couple set up due to F_{12} and F_{32}

$$= 3.06 \times 115 = 352 \text{ kN-mm} = 3.52 \text{ kN-m counter-clockwise}$$

Required torque = 3.52 kN-m clockwise

ANALYTICAL METHOD

Refer the figure in the question.

For a slider crank mechanism, α can be found as

$$\cos \alpha = \frac{1}{n} \sqrt{n^2 - \sin^2 \theta}$$

where $n = \frac{l}{r} = 4.5$ and $\theta = 120^\circ$

$$\cos \alpha = 0.981 \quad \therefore \alpha = 11.1^\circ$$

Considering the equilibrium of link 4 by resolving forces into vertical and horizontal components,

$$\Sigma F_H = 0 \quad \text{and} \quad \Sigma F_V = 0$$

$$\therefore F_{34} \cos 11.1 = 3 \text{ kN and } F_{34} = 3.04 \text{ kN}$$

$$F_{34} \sin 11.1 = F_{14}$$

$$\therefore F_{14} = 0.59 \text{ kN}$$

$$\angle OAB = 180 - (120 + \alpha) = 180 - 131.1 = 48.9^\circ$$

$$h = OA \sin 48.9 = 150 \sin 48.9 = 113 \text{ mm}$$

$$\therefore T = F_{34} \times h = 3.06 \times 113 = 345.78 \text{ N-mm} = \mathbf{3.45 \text{ kN-m clockwise}}$$

PROBLEM 1.2. *A four bar linkage has links of the given dimensions, subject to forces specified in the table. For the configuration when the crank AB makes 60° with the fixed link, determine*

- Torque T on AB to maintain the linkage in static equilibrium*
- Forces on the bearings at A, B, C and D.*

Link	Length, mm	Force	Magnitude, N	Pt. of application
Crank AB	660	F_1	$106 \angle 75^\circ$	430 mm from A
Coupler BC	872	F_2	$190 \angle 60^\circ$	393 mm from A
O/P Link CD	740	F_3	$80 \angle 45^\circ$	492 mm from A
Fixed Link AD	1320	—	—	—

Chapter 2

Governors

A governor is a mechanical device used to maintain the speed of an engine or other prime mover within specified limits irrespective of the load acting on it.

In an engine, speed variations can occur either within a cycle of operation or over a number of cycles. In the first case, the cyclic variation occurs due to difference in the output torque during different strokes of the engine. This can be corrected using a flywheel.

In the second case, speed variation occurs due to the variation in load acting on the engine. When the load on an engine increases, its speed tends to decrease and vice versa. A governor ensures that a more or less constant speed is maintained at different loads by regulating the fuel supply. At high loads, more fuel is supplied by opening the throttle valve and when loads are low the fuel supply is cut off or reduced.

2.1

Classification of governors

On the basis of their working principle, governors can be classified as centrifugal and inertia governors. Of these, centrifugal governors are more commonly used.

Centrifugal governors

A centrifugal governor consists of two masses or balls connected to arms as shown in fig. 1.

The upper arms are fixed on the spindle or shaft while the lower arms are connected to a sleeve which is free to slide on the spindle between two stops. When the



FIGURE 1. Centrifugal governor - structure and working

engine rotates, the balls too rotate. As the speed increases, the radius of rotation of the balls too increase and they raise the sleeve upwards along the shaft. If the speed is decreased the radius of rotation decreases and the sleeve is pushed downwards. This sleeve is connected to a lever which controls the throttle valve. When the sleeve moves up, the valve closes and cuts off the fuel supply thereby reducing engine speed. If the sleeve moves downwards, the valve opens and more fuel is supplied, increasing the engine speed.

The different types of centrifugal governors are shown in the chart below.

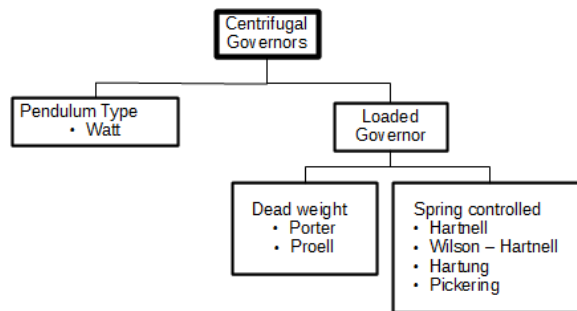


FIGURE 2. Types of governors

Inertia governor

While a centrifugal governor works using the change in speed of an engine, the inertia governor uses the rate of change of engine speed for acting. Inertia governor consists of an arm AB with a massive ball m and a spring CD mounted on a disc keyed to the engine shaft.



FIGURE 3. Inertia governor - structure and working

Initially, when the engine works at a particular speed ω_1 , the position of m is as shown in fig.3. CD constrains the displacement of AB. When the engine speed increases, the disc accelerates to ω_2 . However, due to inertia, m moves in the opposite direction as shown. This movement is used to work the throttle valve and the fuel supply is reduced until the equilibrium position is reached.

Inertia governors have lower response time than centrifugal governors, but their balancing is very difficult. Due to their simplicity centrifugal governors are more popular than inertia governors.

2.2

Terms used in the analysis of governors

Some of the key terms used in the study of governors are defined below

Height of the governor - Height of the governor h is defined as the vertical distance between the centre of the balls and the point where the arms of the governor intersect the spindle axis.

Radius of rotation - Radius of rotation r is the horizontal distance between the centre of a ball and the axis of rotation of the governor.

Equilibrium Speed - A speed of rotation at which there is no upward or downward movement of the governor sleeve is called equilibrium speed. This implies that the parts of the governor - balls, sleeve, links and arms are in complete equilibrium at this speed.

Mean equilibrium speed - The equilibrium speed corresponding to mean position of the governor is called mean equilibrium speed.

Maximum and minimum equilibrium speeds - The equilibrium speeds corresponding to maximum and minimum radii of rotation are called maximum and minimum equilibrium speeds respectively.

Sleeve lift - The vertical distance through which the sleeve moves for a certain change in equilibrium speed is called sleeve lift.

Note : *For every possible radius of rotation, there is a corresponding equilibrium speed. Hence there is an infinite number of possible equilibrium speeds between the maximum and minimum positions.*

2.3

Watt Governor

Watt governor is a pendulum type centrifugal governor, also known as a simple conical governor. It is the oldest and simplest type of governor. It works satisfactorily for speeds ranging from 60 to 80 rpm.

Depending on how the spindle arms are joined to the axis, Watt governors can be classified as shown in figure 4



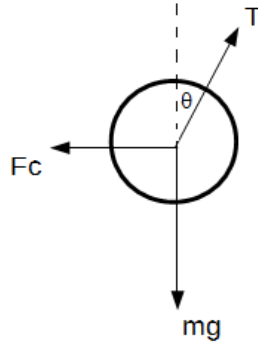
FIGURE 4. Types of Watt governor

When the speed of the engine increases, the balls get lifted and the radius of rotation increases. This will pull the sleeve up along the shaft axis closing the throttle valve and thus reducing the speed by cutting off the fuel supply. If the engine slows down, the balls get lowered and the radius of rotation decreases. This will push the sleeve down along the shaft axis opening the throttle valve and admitting more fuel to increase the engine speed.

Analysis of Watt governor In the analysis of a Watt governor, it is assumed that the mass of sleeve, arms and links are negligible.

Refer fig. 4(a). Let m be the mass of the ball in kilograms, ω be the angular velocity of the arm about the spindle axis in rad/s, h be the height of the governor and r the radius of rotation in metres and F_c the centrifugal force acting on the balls in Newtons. The free body diagram of a ball of the Watt governor is drawn.

The mass m is in equilibrium under the action of three forces - the weight of ball, mg , the centrifugal force acting on the mass, F_c and the tension in the arm, T . The arm makes an angle θ with the vertical.

FIGURE 5. Free body diagram of mass m

We know that

$$F_c = m\omega^2 r \text{ and } \omega = \frac{2\pi N}{60}$$

= For equilibrium,

$$\Sigma H = \Sigma V = \Sigma M = 0$$

Equating horizontal forces, $T \sin \theta = F_c = m\omega^2 r$

Equating vertical forces, $T \cos \theta = mg$

Dividing,

$$\tan \theta = \frac{r\omega^2}{g} \quad (2.1)$$

From fig. 4(a),

$$\frac{r}{h} = \frac{r\omega^2}{g} \quad (2.2)$$

Taking moments about the point O on the axis,

$$F_c \times h = m.g \times r$$

So,

$$m\omega^2 r \times h = m.g \times r \text{ or } h = \frac{g}{\omega^2}$$

Substituting for ω and $g = 9.81 \text{ m/s}^2$

$$\text{i.e. } h = \frac{9.81}{\left(\frac{2\pi N}{60}\right)^2} = \frac{9.81 \times 60^2}{(2\pi)^2 \times N^2}$$

$$\text{Height of the governor, } h = \frac{895}{N^2} \quad (2.3)$$

From equation 2.3 it is seen that at high speeds, the height of governor becomes small. Small changes in speed are not sufficient to cause enough sleeve movement to work the throttle valve. This can be corrected by adding a weight to the sleeve.

2.4

Solved Problems - Watt Governor

PROBLEM 2.1. *A closed-type Watt governor has upper arms of length 360 mm. Calculate the percentage change in speed when the angle made by the arm with the vertical changes from 30° to 40°*

SOLUTION. From equation 2.1 ,

$$\tan \theta = \frac{r\omega^2}{g}$$

Also,

$$\sin \theta = \frac{r}{360} \quad \text{and} \quad \tan \theta = \frac{r}{h}$$

At $\theta = 30^\circ$,

Radius of rotation, $r = 360 \sin \theta = 180 \text{ mm}$

Height of governor, $h = \frac{180}{\tan 30} = 312 \text{ mm}$

Speed of rotation

$$N_1 = \sqrt{\frac{895000}{h}} = \sqrt{\frac{895000}{312}} = 53.5 \text{ rpm}$$

At $\theta = 40^\circ$,

Radius of rotation, $r = 360 \sin \theta = 231.5 \text{ mm}$

Height of governor, $h = \frac{231.5}{\tan 40} = 276 \text{ mm}$

Speed of rotation

$$N_2 = \sqrt{\frac{895000}{h}} = \sqrt{\frac{895000}{276}} = 57 \text{ rpm}$$

Change in speed, $\Delta N = N_2 - N_1 = 57 - 53.5 = 3.5 \text{ rpm}$

Percentage change in speed = $\frac{\Delta N}{N_1} \times 100 = \frac{3.5}{53.5} \times 100 = \mathbf{6.54 \text{ percent}}$

Determine the sleeve lift when a Watt governor changes its speed of rotation from 50 rpm to 70 rpm.

Solution :

From equation 2.3,

$$h = \frac{895}{N^2}$$

At $N_1 = 50 \text{ rpm}$,

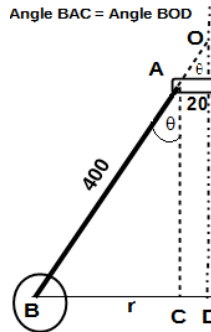
$$h_1 = \frac{895000}{50^2} = 358 \text{ mm}$$

At $N_2 = 70 \text{ rpm}$,

$$h_2 = \frac{895000}{70^2} = 182.65 \text{ mm}$$

Sleeve lift, $\Delta h = h_1 - h_2 = 175.35 \text{ mm} \approx \mathbf{175 \text{ mm}}$

PROBLEM 2.2. An open arm Watt governor has a top horizontal link of length 40 mm and arms of length 400 mm. Initially, the arm makes an angle of 30° with the vertical. Determine the increase in speed for a sleeve lift of 30 mm.



SOLUTION.

FIGURE 6. Problem 3

Refer fig. 6 and consider $\triangle ABC$.

$$\sin 30 = \frac{(r - 30)}{400} \Rightarrow (r - 30) = 200 \text{ mm} \Rightarrow r = 230 \text{ mm}$$

Now consider $\triangle BOD$

$$\tan 30 = \frac{h}{r} = \frac{h}{230}$$

$$\text{Initial governor height} = h_1 = 230 \tan 30 = 133 \text{ mm}$$

$$\text{Initial speed} = N_1 = \sqrt{\frac{895000}{h_1}} = 82 \text{ rpm}$$

$$\text{Sleeve lift} = h_1 - h_2 = 30 \text{ mm} \Rightarrow \text{Final height, } h_2 = 113 \text{ mm}$$

$$\text{Final Speed, } N_2 = \sqrt{\frac{895000}{h_2}} = 89 \text{ rpm}$$

Increase in speed = **7 rpm**

Determine the height of the governor and radius of rotation for a Watt governor having arms of length $l = 300$ mm, rotating at a speed of 75 rpm.

Solution :

Length of arm, $l = 300$ mm

Speed of rotation, $N = 75$ rpm

Governor height

$$h = \frac{895000}{N^2} = 159.11 \text{ mm} \approx 160 \text{ mm}$$

At $N = 75$ rpm,

$$\theta = \cos^{-1} \frac{h}{l} = \cos^{-1} \frac{160}{300} = 58^\circ$$

$$\tan \theta = \frac{r\omega^2}{g}$$

$$\text{Hence, radius of rotation} = \frac{g \tan 58^\circ}{\left(\frac{2\pi N}{60}\right)^2} = \frac{60 \times 60 \times 9.81 \times \tan 58^\circ}{(2\pi \times 75)^2} = 0.254 \text{ m} = 254 \text{ mm}$$

2.5

Porter governor

Porter governor is a modified Watt governor in which a central load is added to the sleeve that slides on the spindle. This added load ensures sufficient sleeve lift at high speeds, and so a Porter governor is suitable for high-speed applications. The construction of a Porter governor is shown in fig 7.



FIGURE 7. Porter governor

Analysis of Porter governor Consider the Porter governor in fig.7.

Let m be the mass of the fly ball, kg

M - mass of sleeve, kg

l - length of governor arm, m

r - radius of rotation, m

h - height of governor, h

From fig ??, the forces acting on the governor are

- (1) Weight of fly ball, mg
- (2) Centrifugal force on flyball, $F_c = mr\omega^2$
- (3) Weight of sleeve, $\frac{Mg}{2}$

The point I is the instantaneous center of rotation of link AB.

Taking moments about I,

$$F_c \times AD = mg \times ID + \frac{Mg}{2} IB$$

(Since $IB = ID + BD$)

$$F_c = mg \times \frac{ID}{AD} + \frac{Mg}{2} \left[\frac{ID}{AD} + \frac{BD}{AD} \right]$$

Consider $\triangle AID$ and $\triangle DAB$

$$\frac{ID}{AD} = \tan \alpha, \quad \frac{BD}{AD} = \tan \beta$$

$$mr\omega^2 = mg \tan \alpha + \frac{Mg}{2} [\tan \alpha + \tan \beta]$$

If $\frac{\tan \beta}{\tan \alpha} = p$ and from $\triangle AOC$, $\tan \alpha = \frac{r}{h}$

$$m\omega^2 = \frac{1}{h} \left[mg + \frac{Mg}{2} (1 + p) \right]$$

$$\omega^2 = \left(\frac{2\pi N}{60} \right)^2 = \frac{g}{h} \left[1 + \frac{M}{2m} (1 + p) \right]$$

From this equation,

$$N^2 = \frac{60^2 \times 9.81}{4\pi^2 h} \left[1 + \frac{M}{2m} (1 + p) \right]$$

On simplification

$$N^2 = \frac{895}{h} \left[1 + \frac{M}{2m} (1 + p) \right] \quad (2.4)$$

If the arms and links of the governor have equal length and are pinned so that O and B are on the same line, $\alpha = \beta$ and $p = 1$. Then equation 2.4 is modified to

$$N^2 = \frac{895}{h} \left[1 + \frac{M}{m} \right] \quad (2.5)$$

Effect of friction In a Porter governor, frictional force which acts in a direction opposite to that of motion may also be considered. If a frictional force F_R also acts on the sleeve and $\alpha = \beta$, the equation gets modified as

$$N^2 = \frac{895}{h} \left[1 + \frac{M + \left(\frac{F_R}{g} \right)}{m} \right] \quad (2.6)$$

→ when sleeve moves upwards

$$N^2 = \frac{895}{h} \left[1 + \frac{M - \left(\frac{F_R}{g} \right)}{m} \right] \quad (2.7)$$

→ when sleeve moves downwards

If $\alpha \neq \beta$, the equation can be written as

$$N^2 = \frac{895}{h} \left[1 + \frac{M \pm \left(\frac{F_R}{g} \right)}{2m} (1 + p) \right] \quad (2.8)$$

depending on the direction of motion of the sleeve.

Note:

- F_R must be in Newtons to use directly in equations 2.6, 2.7 and 2.8.
- When sleeve moves up, F_R acts downwards adding to the weight Mg of the sleeve. Hence equation 2.6 is appropriate.
- When sleeve moves down, F_R acts upwards opposing the weight Mg of the sleeve. Hence equation 2.7 is appropriate.
- Between the two values of N given by equation 2.8, the governor will be insensitive.

2.6 _____

Proell Governor

2.7 _____

Hartnell Governor

Chapter 3

Flywheels

A flywheel is a massive wheel which is keyed on to an engine shaft for the purpose of regulating the speed of the engine.

In a four-stroke engine, energy is produced only during the working stroke. During the suction, compression and exhaust strokes, no energy is produced. A flywheel stores the energy produced during the power or working stroke and distributes it during the other three strokes so that the speed of the engine does not vary significantly.

Applications : IC Engines, punching machines, rivetting machines etc.

3.1

Turning Moment Diagram

A graph showing turning moment vs. crank angle is called turning moment diagram or $T - \theta$ diagram. It represents the engine torque or turning moment for different positions of the crank. The area under the turning moment diagram represents the work done.

For a single cylinder engine, the turning moment is given by

$$T = F_P \times r \left(\sin\theta + \frac{\sin 2\theta}{2\sqrt{n^2 - \sin^2\theta}} \right) \quad (3.1)$$

F_P = piston effort

r = crank radius/crank length

θ = angle turned by the crank from inner dead center position

$n = \frac{l}{r}$ = length of connecting rod divided by crank radius



FIGURE 1. Turning moment diagram of 4-stroke engine

For a four-stroke engine, the turning moment diagram is as shown in figure 8.

Suction Stroke - $\theta = 0$ to 2π radians - Initially work is required to move the piston and a negative loop is obtained. Later when incoming gases do work on the piston, the loop becomes positive.

Compression Stroke - $\theta = \pi$ to 2π radians - Work is done on the piston to compress the gases. Therefore, a fully negative loop is obtained.

Power Stroke - $\theta = 2\pi$ to 3π radians - The combustion in the engine cylinder causes gases to expand and pushes the piston from TDC to BDC. Work is done by the piston, and a positive loop is drawn.

Exhaust Stroke - $\theta = 3\pi$ to 4π radians - Piston moves back to the top position, making the loop negative. The effect of inertia of the piston during the second part of exhaust stroke makes the loop positive.



FIGURE 2. Turning moment diagram of multicylinder engine

From this diagram, we can see that the work done is maximum during the power stroke,

In a multi-cylinder engine, due to the phase difference between the cylinders, at least one cylinder will produce energy and undergo working stroke at any instant. So there will be less fluctuation in turning moment as we can see in figure 2. When number of cylinders increases, the fluctuation decreases.

Mean Resisting Torque, T_{mean} Mean resisting torque is defined as the work required to overcome the load on the engine. If the torque exceeds T_{mean} , the engine will accelerate, and if the torque falls below T_{mean} the engine will decelerate.

3.2

Working of Flywheel

Flywheel is mounted on the crankshaft, of an engine. The idealized turning moment diagram of an engine is shown in figure 2.



As the crank angle increases from O to A, the loop is negative and there is a shortage of energy causing energy to slow down. From A to B, there is an excess of energy which should cause the engine to accelerate. The massive flywheel stores this excess energy and supplies it back to the engine during the negative portions of the cycle.

3.3

Difference between Flywheel and Governor

Though both flywheel and governor control speed, they have differences in their working and function. These are summarized in the table below

Flywheel	Governor
1. Works continuously whenever engine is in operation	1. Acts only when there is variation in speed
2. Works to control speed variation due to fluctuation of turning moment within a single cycle	2. Controls change in speed over a number of cycles
3. Working is independent of load on engine	3. Responds to speed variations due to change in load on engine
4. Does not maintain constant speed. Only reduces the fluctuation of speed	4. Maintains mean speed of the engine when load on engine varies
5. Works by absorbing and releasing excess energy during engine cycles	5. Maintains engine speed by regulating fuel supply



FIGURE 3. Turning moment diagram of multi-cylinder engine

3.4

Fluctuation of Energy and Speed

The change in energy and speed can be obtained from the turning moment diagram of an engine. Consider the turning moment diagram of a multi-cylinder engine as shown in figure 3.

The area of the loops represent the change in energy. Areas below the mean resisting torque line are taken as negative and those above are taken as positive.

Let E_1 be the energy at point 1

$$E_2 = E_1 + A_1$$

$$E_3 = E_2 - A_2 = E_1 + A_1 - A_2$$

$$E_4 = E_3 + A_3 = E_1 + A_1 - A_2 + A_3$$

$$E_5 = E_4 - A_4 = E_1 + A_1 - A_2 + A_3 - A_4$$

Since the cycle is completed at 5, $E_1 = E_5$.

If E_{max} is the maximum energy value that occurs in a cycle and E_{min} is the minimum energy value, then

Maximum fluctuation of energy, $\Delta E = E_{max} - E_{min}$

If N_{max} and N_{min} are the speeds corresponding to E_{max} and E_{min} ,

Maximum fluctuation of speed, $\Delta N = N_{max} - N_{min}$

Coefficient of fluctuation of speed It is the ratio between the maximum fluctuation of speed and the mean speed.

$$C_N = \frac{\Delta N}{N} = \frac{N_{max} - N_{min}}{N} \quad (3.2)$$

where mean speed

$$N = \frac{N_{max} + N_{min}}{2}$$

$$\therefore C_N = \frac{2(N_{max} - N_{min})}{N_{max} + N_{min}} \quad (3.3)$$

In terms of angular velocity ω ,

$$C_N = \frac{2(\omega_{max} - \omega_{min})}{\omega_{max} + \omega_{min}} \quad (3.4)$$

Coefficient of Steadiness It is the reciprocal of the coefficient of fluctuation of speed, and is denoted by m .

$$m = \frac{1}{C_N} = \frac{N_{max} + N_{min}}{2(N_{max} - N_{min})} \quad (3.5)$$

Coefficient of Fluctuation of Energy It is the ratio of the maximum fluctuation of energy ΔE to the work done in a cycle.

$$C_E = \frac{\Delta E}{W} = \frac{E_{max} - E_{min}}{W} \quad (3.6)$$

$$W = T_{mean} \times \theta$$

where,

θ = crank angle per cycle in radians, 2π for two stroke IC engines and 4π for four stroke IC engines

or

$$W = \frac{P \times 60}{n}$$

where,

P = Power transmitted in Watts

n = No. of working strokes / minute, N for two stroke engines and $\frac{N}{2}$ for four stroke engines.

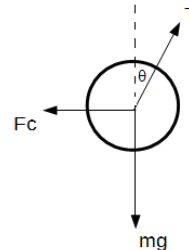
3.5

Energy stored in a flywheel

A flywheel is usually massive, with most of its mass placed close to the rim.

When the flywheel absorbs energy, its speed increases. When it dispenses energy, its speed decreases.

Consider a flywheel of mass m kilograms. Let I be the moment of inertia about its axis of rotation such that $I = mk^2$ where k is the radius of gyration in metres.



The kinetic energy of a rotating fly-wheel

$$E = \frac{1}{2}I\omega^2 = \frac{1}{2}mk^2\omega^2$$

If ω_{max} and ω_{min} are the maximum and minimum angular velocities attained during a cycle, the maximum fluctuation of energy

$$\begin{aligned}\Delta E &= \frac{1}{2}I\omega_{max}^2 - \frac{1}{2}I\omega_{min}^2 \\ &= \frac{1}{2}I(\omega_{max}^2 - \omega_{min}^2) \\ &= \frac{1}{2}I(\omega_{max} + \omega_{min})(\omega_{max} - \omega_{min}) \\ &= I\omega(\omega_{max} - \omega_{min})\end{aligned}$$

Divide and multiply by ω

$$\Delta E = I\omega^2 \frac{(\omega_{max} - \omega_{min})}{\omega} = I\omega^2 C_N = mk^2\omega^2 C_N \quad (3.7)$$

Alternately,

$$\Delta E = 2EC_N \quad \text{where } E = \frac{1}{2}I\omega^2 \quad (3.8)$$

If radius of gyration k is taken as equal to the mean radius R of the flywheel,

$$\Delta E = mk^2\omega^2 C_N = mR^2\omega^2 C_N = mv^2 C_N \quad (3.9)$$

where $\omega r = v =$ linear velocity of flywheel at mean radius R in m/s.

In terms of revolutions per minute, N

$$\Delta E = \frac{\pi^2}{900}mk^2N^2C_N \quad (3.10)$$

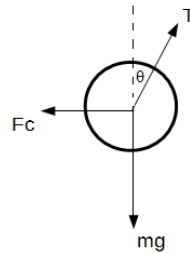
3.6

Design of Flywheel Rim

Consider a flywheel of mass m having a rim of breadth b and thickness t rotating at a speed of N rpm.

When it rotates, a centrifugal force F_C acts on the rim. Consider an element at angle θ subtending an angle $d\theta$ at the centre as shown in figure 4. Let dF_C be the centrifugal force acting on this element.

The total vertical force acting on half the rim can



be obtained as

$$F_{CV} = \int_0^\pi dF_C \sin\theta$$

But centrifugal force

$$dF_C = mr\omega^2$$

$$\begin{aligned} & \text{since the mass } m = \rho R d\theta A \text{ where } A \text{ is the cross-sectional area of the rim} \\ & = \rho \times R d\theta \times A \times R\omega^2 \\ & = \rho R^2 A \omega^2 d\theta \end{aligned}$$

$$\begin{aligned} \therefore F_{CV} &= \int_0^\pi \rho R^2 A \omega^2 \sin\theta d\theta \\ &= \rho R^2 A \omega^2 [-\cos\theta]_0^\pi \\ &= \rho R^2 A \omega^2 [1 - (-1)] \\ &= 2\rho R^2 A \omega^2 \end{aligned}$$

This force will be resisted by the hoop stress σ such that

$$\begin{aligned} 2 \times \sigma \times b \times t &= 2\rho R^2 A \omega^2 \\ \therefore \sigma &= \rho R^2 \omega^2 = \rho v^2 \\ \therefore v &= \sqrt{\frac{\sigma}{\rho}} \end{aligned} \tag{3.11}$$

Also, the mass $m = 2\pi R \times A \times \rho$

$$\therefore A = \frac{m}{2\pi R \rho} = b \times t \tag{3.12}$$

The flywheel rim can be designed on the basis of the equations 3.11 and 3.12.



FIGURE 4. Forces on flywheel rim element

3.7

Flywheel in Punching Press

In an engine, a flywheel is used to supply constant output torque while the input torque from the engine varies. In a machine like a punching press or riveting machine, the input torque from the motor is constant while the flywheel ensures varying output torque depending on the load.

Punching press is used to punch holes in metal sheets. The load on the press is maximum during the punching operation itself i.e. from θ_1 to θ_2 . For the rest of the cycle, there is zero load.



FIGURE 5. Punching press

If the energy required for punching one hole is E , then the energy supplied by motor during actual punching,

$$E_m = E \left(\frac{\theta_2 - \theta_1}{2\pi} \right) \quad (3.13)$$

Since the entire energy E is required during the punching operation alone, the flywheel must supply the deficit energy.

The maximum fluctuation of energy when flywheel velocity changes from ω_1 to ω_2

$$\Delta E = E_f = E \left[1 - \frac{\theta_2 - \theta_1}{2\pi} \right] \quad (3.14)$$

But

$$E = \frac{1}{2} \times F_S \times t$$

where,

$F_S = \pi d t \sigma_s$ Maximum shear force

σ_s = ultimate shear stress, d = for hole diameter and t = sheet thickness.

It is also assumed that

$$\frac{\theta_2 - \theta_1}{2\pi} \approx \frac{t}{2s} = \frac{t}{4r}$$

where S = stroke of punch = $2 \times$ crank radius, r

3.8

Solved Problems

PROBLEM 3.1. *An engine has a flywheel of mass 5000 kg and radius of gyration 1.5 m. The maximum engine speed is 130 rpm while the minimum is 124 rpm. Find the maximum fluctuation of energy.*

SOLUTION. Given : $m = 5000 \text{ kg}$ $k = 1.5 \text{ m}$

$N_{max} = 130 \text{ rpm}$ $N_{min} = 124 \text{ rpm}$

$$\therefore \text{Mean Speed, } N = \frac{N_{max} + N_{min}}{2} = \frac{130 + 124}{2} = 127 \text{ rpm}$$

Coefficient of fluctuation of speed

$$C_N = \frac{N_{max} - N_{min}}{N} = \frac{130 - 124}{127} = 0.047 = 4.7 \text{ percent}$$

Maximum fluctuation of energy

$$\Delta E = \frac{\pi^2}{900} m k^2 N^2 C_N = \frac{\pi^2}{900} \times 5000 \times 1.5^2 \times 127^2 \times 0.047 = \mathbf{93.52 \text{ kN.m}}$$

PROBLEM 3.2. *The maximum fluctuation of energy in a flywheel of 6 tonne mass is 50 kN-m. If the radius of gyration is 1.2 m and the mean speed is 120 rpm, find the maximum and minimum speeds*

SOLUTION. Given : $\Delta E = 50 \times 10^3 \text{ N-m}$ $m = 6000 \text{ kg}$ $k = 1.2 \text{ m}$ $N = 120 \text{ rpm}$

$$\Delta E = \frac{\pi^2}{900} m k^2 N^2 C_N$$

$$50 \times 10^3 = \frac{\pi^2}{900} \times 6000 \times 1.2^2 \times 120^2 \times C_N$$

$\therefore$ Coefficient of fluctuation of speed

$$C_N = \frac{N_{max} - N_{min}}{120} = 0.0366$$

$$\therefore N_{max} - N_{min} = 4.398$$

Also, the mean speed

$$N = \frac{N_{max} + N_{min}}{2} = 120 \quad \therefore N_{max} + N_{min} = 240$$

From the above, $2N_{max} = 244.398 \approx 244.5$

$$N_{max} = 122.25 \text{ rpm} \approx \mathbf{122 \text{ rpm}}$$

$$N_{min} = 117.75 \text{ rpm} \approx \mathbf{118 \text{ rpm}}$$

PROBLEM 3.3. *A 4-stroke IC engine develops 100 kW power. The maximum and minimum speeds are within ± 1.5 percent of the mean speed of 225 rpm. Determine the flywheel mass required if the maximum fluctuation of energy is constrained to 25 percent of the work done per stroke. Assume radius of gyration $k = 0.75 \text{ m}$*

SOLUTION. Given : Power developed = 100 kW $N = 225 \text{ rpm}$ $k = 0.75 \text{ m}$

$$T_{mean} = \frac{P}{\omega} = \frac{P \times 60}{2\pi N} = \frac{100 \times 10^3 \times 60}{2\pi \times 225} = 4244 \text{ N-m}$$

$$\text{Work done per cycle} = T_{mean} \times \theta = 53333.3 \text{ N-m},$$

where $\theta = 4\pi$ for 4-stroke engine.

Work done per stroke =

$$\frac{\text{W.D. per cycle}}{\text{No. of strokes}} = \frac{53333.3}{4} = 13333.3 \text{ N-m}$$

Maximum allowable fluctuation of energy $\Delta E = 0.25 \times 13333.3 = 3333 \text{ N-m}$

But

$$\Delta E = \frac{\pi^2}{900} m k^2 N^2 C_N$$

Since the speed can vary only by 1.5 percent on either side of the mean speed, N

$$C_N = 0.03$$

Substituting in the equation for ΔE ,

$$3333 = \frac{\pi^2}{900} \times m \times 0.75^2 \times 225^2 \times 0.03$$

$\therefore$ Mass of required flywheel, $m = 355.76 \text{ kg} \approx \mathbf{356 \text{ kg}}$

PROBLEM 3.4. *For the turning moment diagram of a multi-cylinder engine given below, the scale is 1 mm = 500 N-m on the y-axis and 1 mm = 5 degrees on the x-axis. The areas in mm^2 are*

$$\begin{array}{lll} A1 = 30 & A2 = 380 & A3 = 250 \\ A4 = 300 & A5 = 320 & A6 = 250 \\ A7 = 375 & A8 = 275 & A9 = 230 \end{array}$$

The mean speed of the engine is 400 rpm and the coefficient of fluctuation of speed is 0.03. If the radius of gyration is 1 m, determine the mass of the flywheel.



SOLUTION. Given : $N = 400 \text{ rpm}$ $C_N = 0.03$ $k = 1 \text{ m}$

From the given scale, 1 mm^2 on the turning moment diagram is equivalent to

$$500 \times \left(5 \times \frac{\pi}{180} \right) = 13.9\pi \text{ N-m}$$

Areas below T_{mean} line are negative and those above are positive.

If E_0 is the energy at 0,

$$E_1 = E_0 - A_1 = E_0 - 30$$

$$E_2 = E_0 - 30 + 380 = E_0 + 350$$

$$E_3 = E_0 + 350 - 250 = E_0 + 100$$

$$E_4 = E_0 + 100 + 300 = E_0 + 400$$

$$E_5 = E_0 + 400 - 320 = E_0 + 80$$

$$E_6 = E_0 + 80 + 250 = E_0 + 330$$

$$E_7 = E_0 + 330 - 375 = E_0 - 45$$

$$E_8 = E_0 - 45 + 275 = E_0 + 230$$

$$E_9 = E_0 + 230 - 230 = E_0 = \text{Energy at } 0$$

From all these energy values, the extreme values are

$$E_{\text{max}} = E_4 = E_0 + 400$$

$$E_{\text{min}} = E_7 = E_0 - 45$$

$$\begin{aligned}
 \therefore \Delta E &= E_{max} - E_{min} \\
 &= E_0 + 400 - (E_0 - 45) \\
 &= 445 \text{ mm}^2 = 445 \times 13.9\pi = 19432 \text{ N-m}
 \end{aligned}$$

$$\text{Now, } \Delta E = \frac{\pi^2}{900} m k^2 N^2 C_N$$

Mean angular velocity,

$$\omega = \frac{2\pi N}{60} = \frac{2\pi \times 400}{60} = 41.9 \text{ rad/s}$$

$$\text{i.e. } 19432 = m \times 1^2 \times (41.9)^2 \times 0.03$$

Required flywheel mass, $m = 368.95 \text{ kg} \approx 370 \text{ kg}$

PROBLEM 3.5. *The torque requirements of a machine during its cycle of operation is given below.*

- *During the first one revolution, the torque rises uniformly from 600 N-m to 3500 N-m*
- *During the next half revolution, it remains constant*
- *In the next half revolution, the torque decreases uniformly to 2000 N-m, and remains constant for one revolution*
- *In the fourth revolution, the torque decreases uniformly to 600 N-m during the first half and remains constant in the second half*

The cycle is completed in four revolutions.

The motor supplying power to this machine rotates at a mean speed of 300 rpm and supplies a constant torque. If the attached flywheel has a mass of 1000 kg and radius of gyration 750 mm, determine the power of the motor and the coefficient of fluctuation of speed.

SOLUTION. *From the question, the turning moment diagram can be drawn as above*



From turning moment diagram,

$$\begin{aligned}
& \text{Total work done per cycle} = \text{Area of turning moment diagram} \\
& = O_3F_2G - (O_1O_3A + BCC_1 + CC_1F_1F_2 + DEE_1 + EE_1F_1F) \\
& = 3500 \times 8\pi - \left[\frac{1}{2} \times 2\pi \times 2900 + \frac{1}{2} \times \pi \times 1500 + 4\pi \times 1500 + \frac{1}{2} \times \pi \times 1400 + \pi \times 1400 \right] \\
& = 16250\pi N - m
\end{aligned}$$

$$\therefore \text{Mean Torque } T_{\text{mean}} = \frac{W.D}{\theta} = \frac{16250\pi}{8\pi} = 2031 \text{ N} - m$$

$$\begin{aligned}
\text{Power of the motor} &= T_{\text{mean}} \times \omega \\
&= T_{\text{mean}} \times \frac{2\pi N}{60} = 2031 \times \frac{2\pi \times 300}{60} \\
&= \mathbf{63805.75 \text{ W} = 63.8 \text{ kW}}
\end{aligned}$$

If the points of intersection of the mean torque line and turning moment diagram are marked as X and Y, then the energy

$$E_Y = E_X + \text{Area above the } T_{\text{mean}} \text{ line}$$

Thus maximum fluctuation of energy $= E_Y - E_X = \text{Area above the mean torque line.}$

$$\therefore \Delta E = \text{Area of trapezium } ABYX = \frac{1}{2}h(AB + XY)$$

Consider $\triangle AO_1A''$ and $\triangle AXA'$ which are similar

$$\begin{aligned}
\frac{A'X}{A''O_1} &= \frac{AA'}{AA''} \\
\therefore \frac{A'X}{2\pi} &= \frac{(3500 - 2031)}{2900} \\
\therefore A'X &= 3.1828 = 3.18
\end{aligned}$$

Consider $\triangle B''BC$ and $\triangle B'BY$

$$\begin{aligned}
\frac{B'Y}{B''C} &= \frac{BB'}{BB''} \\
\therefore \frac{B'Y}{(4\pi - 3\pi)} &= \frac{(3500 - 2031)}{(3500 - 2000)} \\
\therefore B'Y &= 3.08
\end{aligned}$$

$$\text{Now, } XY = A'X + A'B' + B'Y = 3.18 + 3.14 + 3.08 = 9.4$$

$$AB = A'B' = 3.14 \text{ and } h = 3500 - 2031 = 1469 \text{ N-m}$$

$$\text{Area of } XABY = \frac{1}{2} \times 1469 (9.4 + 3.14) = 9210 \text{ N-m}$$

$$\begin{aligned}\Delta E &= mk^2 \omega^2 C_S = 9210 \text{ N-m} \Rightarrow 9210 = 1000 \times 0.75^2 \times 31.4^2 \times C_S \\ \therefore C_S &= 0.0166 = \mathbf{1.66 \text{ percent}}\end{aligned}$$

PROBLEM 3.6. *The turning moment diagram of a two-stroke engine varies with respect to crank angle θ as*

$$T = 30000 - 9000 \sin 3\theta \text{ N-m}$$

If the engine has a mean speed of 320 rpm and has a flywheel of moment of inertia 300 kg-m², calculate the engine power and coefficient of fluctuation of speed if the resisting torque is constant.

$$\begin{aligned}\text{SOLUTION. Given : } T &= 30000 - 9000 \sin 3\theta \text{ N-m} \\ N &= 320 \text{ rpm} \quad I = 300 \text{ kg-m}^2\end{aligned}$$

$$\omega = 33.5 \text{ rad/s}$$

For a two-stroke engine, the working cycle takes 360°. But torque T is given as a function of 3θ .

$$3\theta = 360^\circ \quad \therefore \theta = 120^\circ$$

The torque cycle repeats after each 120° or $\frac{2\pi}{3}$ radians.

Work done per cycle

$$\begin{aligned}&= \int_0^{\frac{2\pi}{3}} T d\theta \\ &= \int_0^{\frac{2\pi}{3}} (30000 - 9000 \sin 3\theta) d\theta \\ &= \left[30000 \theta - \frac{9000 \cos 3\theta}{3} \right]_0^{\frac{2\pi}{3}} \\ &= 20000 \pi \text{ N-m}\end{aligned}$$

$$\text{Mean resisting torque} = T_{\text{mean}} = \frac{\text{W.D. per cycle}}{\text{Crank angle/cycle}} = \frac{20000\pi}{2\pi \cdot 3} = 30000 \text{ N-m}$$

Power developed by the engine =

$$T_{\text{mean}} \times \omega = 30000 \times 33.5 = 1005000 \text{ W} = \mathbf{1005 \text{ kW}}$$

At any instant,

$$\Delta T = T - T_{\text{mean}} = (30000 - 9000 \sin 3\theta) - 30000 = -9000 \sin \theta$$

Also, $\Delta T = 0$ when $T - T_{\text{mean}} = -9000 \sin 3\theta = 0$

$$\sin 3\theta = 0 \Rightarrow 3\theta = 0^\circ \text{ or } 180^\circ$$

$$\therefore \theta = 0^\circ \text{ or } 60^\circ$$

The shape of the curve is sinusoidal but negative.

The turning moment diagram can be drawn as follows to determine the maximum fluctuation of energy.



. Maximum fluctuation of energy occurs between 0° and 60°

$$\begin{aligned} \therefore \Delta E &= \int_{\theta_0}^{\theta_1} \Delta T \cdot d\theta = \int_{60^\circ}^{0^\circ} (-9000 \sin 3\theta) d\theta \\ &= \frac{9000 \cos 3\theta}{3} \Big|_{0^\circ}^{60^\circ} = -3000 - 3000 = -6000 \text{ N-m} \end{aligned}$$

Also,

$$\Delta E = I\omega^2 C_N \Rightarrow 6000 = 300 \times 33.5^2 \times C_N$$

Coefficient of fluctuation of speed, $C_N = 0.0178 \approx 1.75 \text{ percent}$

PROBLEM 3.7. For the flywheel in Problem 6, determine the coefficient of fluctuation of speed if the mean resisting torque is given as

$$T_{\text{mean}} = 30000 + 4500 \sin \theta \text{ N-m}$$

Also determine the maximum and minimum angular acceleration.

SOLUTION. We need to first draw the turning moment diagram corresponding to this case.

$$\text{Given } T_{\text{mean}} = 3000 + 4500 \sin \theta$$

At any instant,

$$\Delta T = T - T_{\text{mean}} = 30000 - 9000 \sin 3\theta - 30000 - 4500 \sin \theta = -9000 \sin 3\theta - 4500 \sin \theta$$

To determine the point where $\Delta T = 0$,

$$\Delta T = 0 \Rightarrow -9000 \sin 3\theta - 4500 \sin \theta = 0$$

$$\text{i.e. } -9000 \sin 3\theta = 4500 \sin \theta$$

Since $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$,

$$-9000(3\sin\theta - 4\sin^3\theta) = 4500\sin\theta$$

$$\therefore 36000 \sin^3\theta = 31500\sin\theta$$

$$\sin^2\theta = 0.875$$

$$\therefore \sin\theta = 0.935$$

Hence at $\theta = 69.29^\circ$ and 110.70° the T_{mean} curve and the turning moment curve will intersect as shown in the figure.



. The maximum fluctuation of energy occurs between $\theta = 69.29^\circ$ and 110.70° .

$$\begin{aligned} \therefore \Delta E &= \int_{\theta'_1}^{\theta'_2} \Delta T \, d\theta = \int_{69.29^\circ}^{110.70^\circ} (-9000\sin 3\theta = 4500\sin\theta) \, d\theta \\ &= \left[\frac{9000\cos 3\theta}{3} + 4500\cos\theta \right]_{69.29}^{110.7} \\ &= 1060.66 - (-2651.68) = 3712.3 \, \text{N-m} \end{aligned}$$

Since $\Delta E = I\omega^2 C_N$,

Coefficient of fluctuation of speed,

$$C_N = \frac{3712.3}{300 \times 33.5^2} = 0.0110 = \mathbf{1.10 \, percent}$$

Maximum and Minimum angular acceleration

The change in torque will be maximum or minimum at points of maximum and minimum angular acceleration.

$\therefore$ At α_{max} and α_{min} ,

$$\frac{d(\Delta T)}{d\theta} = 0$$

$$\begin{aligned} \frac{d}{d\theta} (-9000\sin 3\theta = 4500\sin\theta) &= 0 \\ -2700 \cos 3\theta - 4500 \cos \theta &= 0 \end{aligned}$$

$$-3 \cos 3\theta = 5 \cos \theta$$

Since $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$,

$$-12 \cos^3 \theta + 9 \cos \theta = 5 \cos \theta$$

$$12 \cos^3 \theta - 4 \cos \theta = 0$$

$$\therefore \cos \theta = 0.577, -0.577 \text{ or } 0$$

$$\theta = 54.76^\circ, 125.24^\circ \text{ or } 90^\circ$$

Substituting for θ in the expression for ΔT ,

$$(\Delta T)_{54.76^\circ} = -6113.8 \text{ N-m}$$

$$(\Delta T)_{125.24^\circ} = -6113.8 \text{ N-m}$$

$$(\Delta T)_{90^\circ} = 4500 \text{ N-m}$$

$$\therefore (\Delta T)_{\min} = -6113.8 \text{ N-m}$$

$$(\Delta T)_{\max} = 4500 \text{ N-m}$$

Maximum angular acceleration

$$\alpha_{\max} = \frac{(\Delta T)_{\max}}{I} = \frac{4500}{300} = 15 \text{ rad/s}^2$$

Minimum angular acceleration or maximum angular retardation

$$\alpha_{\min} = \frac{(\Delta T)_{\min}}{I} = \frac{-6113.8}{300} = -20.4 \text{ rad/s}^2$$

PROBLEM 3.8. *In the turning moment diagram of a flywheel, the areas above and below the mean torque line in mm² are given as follows*

$$A1 = +500 \quad A2 = -25 \quad A3 = +250 \quad A4 = -340$$

$$A5 = +120 \quad A6 = -250 \quad A7 = +225 \quad A8 = -480$$

Scale: *x-axis* - 1mm = 6°, *y-axis* - 1mm = 1000 N-m

The engine has a mean speed of 180 rpm and the coefficient of fluctuation of speed is limited to 4percent. The hub and spokes of the flywheel account for 5 percent of the rotational inertia. The peripheral velocity of the flywheel is 18 m/s and the density of flywheel material is 7200 kg/m³. Design the flywheel. Assume rim thickness as 30 cm

SOLUTION. *Designing a flywheel involves determination of*

- (1) *Diameter*
- (2) *Mass*
- (3) *Cross-sectional area of the rim*

Given : $N = 180 \text{ rpm}$ $C_N = 0.04$ $\rho = 7200 \text{ kg/m}^3$ $v = 18 \text{ m/s}$

$$\therefore \omega = \frac{2\pi N}{60} = 18.85 \text{ rad/s}$$

From the given data, the turning moment diagram can be sketched



Let energy at 0 = E_0

$$E_1 = E_0 + 500$$

$$E_2 = E_0 + 500 - 25 = E_0 + 475$$

$$E_3 = E_0 + 475 + 250 = E_0 + 725$$

$$E_4 = E_0 + 725 - 340 = E_0 + 385$$

$$E_5 = E_0 + 385 + 120 = E_0 + 505$$

$$E_6 = E_0 + 505 - 250 = E_0 + 255$$

$$E_7 = E_0 + 255 + 225 = E_0 + 480$$

$$E_8 = E_0 + 480 - 480 = E_0$$

$$E_{max} = E_0 + 725$$

$$E_{min} = E_0$$

Maximum fluctuation of energy,

$$\Delta E = E_{max} - E_{min} = (E_0 + 725) - E_0 = 725 \text{ mm}^2$$

From the given scale,

$$1 \text{ mm}^2 = 1000 \times 6 \times \frac{\pi}{180} = 104.72 \text{ N-m}$$

$$\Delta E = 725 \times 104.72 = 75921.8 \text{ N-m}$$

Design of flywheel

(1) Diameter of flywheel rim

Peripheral velocity, $v = 18 \text{ m/s}$

$$18 = \frac{\pi \times D \times 180}{60} \quad \therefore D = 1.91 \text{ m}$$

Mean diameter of flywheel rim = **1.91 m**

(2) *Mass of flywheel rim*

Maximum fluctuation of energy, $\Delta E = E \times 2C_N$

$$75921.8 = E \times 2 \times 0.04$$

$$E = \frac{75921.8}{2 \times 0.04} = 949022.5 \text{ N-m}$$

Since the hub and spokes account for 5 percent of the energy, the energy of the flywheel rim alone is given by

$$E_{rim} = 0.95E = 901571.4 \text{ N-m}$$

Also,

$$E_{rim} = \frac{1}{2}mv^2$$

$$901571.4 = \frac{1}{2} \times m \times 18^2$$

Mass of flywheel rim, $m =$

$$\frac{2 \times 901571.4}{18^2} = 5565.25 \text{ kg} \approx 5565 \text{ kg}$$

(3) *Cross-sectional area of rim*

Area of flywheel rim,

$$A = \frac{m}{2\pi R\rho} = \frac{5565}{2 \times \pi \times 0.955 \times 7200} = 0.1288 \text{ m}^2$$

Also, area = $b \times t$

If thickness is taken as 30 cm,

Breadth,

$$b = \frac{0.1288}{0.3} = 0.429 \text{ m}$$

PROBLEM 3.9. *A flywheel keyed to a punching machine can run at a maximum speed of 300 rpm and has a radius of gyration of 0.4 m. The press can punch 720 holes per hour and each punching operation in itself takes 2 seconds to complete. A work of 9000 N-m is required to punch one hole. If the minimum permissible speed of the flywheel is 270 rpm, determine the power of the motor required and the minimum mass of the flywheel.*

SOLUTION. *Given : $N_{max} = 300 \text{ rpm}$ $N_{min} = 270 \text{ rpm}$ $k = 0.4 \text{ m}$*

Work done/hole = 9000 N-m

Rate of punching = 720 holes per hour

Total work requirement per hour = $9000 \times 720 = 6480000 \text{ N-m/hour}$

Power requirement = Work required per second =

$$\frac{6480000}{3600} = 1800 \text{ N-m/s} = \mathbf{1800 \text{ W} = 1.8 \text{ kW}}$$

Energy required to punch one hole, $E = 9000 \text{ N-m}$

Time required for each punching operation = 2 seconds

Energy supplied by the motor, $E_m = 1800 \times 2 = 3600 \text{ W}$

Energy supplied by the flywheel during punching operation

$$E_f = E - E_m = 9000 - 3600 = 5400 \text{ N-m}$$

Maximum fluctuation of energy

$$\Delta E = E_f = \frac{\pi^2}{900} m k^2 N^2 C_N = 5400 \text{ N-m}$$

Mean speed, $N =$

$$\frac{N_{max} + N_{min}}{2} = \frac{300 + 270}{2} = 285 \text{ rpm}$$

Coefficient of fluctuation of speed,

$$C_N = \frac{N_{max} - N_{min}}{N} = \frac{300 - 270}{285} = .1052$$

$$\therefore 5400 = \frac{\pi^2}{900} m k^2 N^2 C_N = \frac{\pi^2}{900} \times m \times 0.4^2 \times 285^2 \times 0.1052$$

$$\text{Mass of flywheel, } m = \mathbf{360.17 \text{ kg} \approx 360 \text{ kg}}$$

PROBLEM 3.10. A 275 kW motor is used to drive a riveting machine having a flywheel of mass 150 kg and 600 mm radius. The flywheel rotates at 275 rpm before the riveting operation which requires 12000 Joules of energy and takes 1.5 seconds. How fast will the flywheel rotate after riveting? What is the rate at which rivets can be closed? Also determine the cross-sectional dimensions of the flywheel rim if the breadth is taken as twice the thickness and density of flywheel material is 7200 kg/m³

SOLUTION. Given: $P = 2.75 \text{ kW} = 2750 \text{ W}$ $m = 150 \text{ kg}$
 $k = R = 600 \text{ mm} = 0.6 \text{ m}$ $N_{max} = 275 \text{ rpm}$ $E = 12000 \text{ J} = 12000 \text{ N-m}$
 Time taken for riveting = 1.5 seconds
 Energy supplied by motor, $E_m = 2750 \times 1.5 = 4125 \text{ N-m}$
 Energy supplied by flywheel, $E_f = E - E_m = 7875 \text{ N-m}$
 $\therefore$ Maximum fluctuation of energy, $\Delta E = 7875 \text{ N-m}$
 Also,

$$\Delta E = \frac{\pi^2}{900} m k^2 N^2 C_N$$

$$\begin{aligned}
&= \frac{\pi^2}{900} m k^2 \left[\frac{N_{max} + N_{min}}{2} \right]^2 \times \left[\frac{N_{max} - N_{min}}{N_{max} + N_{min}} \right] \times 2 \\
&= \frac{\pi^2}{900} m k^2 \frac{(N_{max}^2 - N_{min}^2)}{2} \\
7875 &= \frac{\pi^2}{900} \times 150 \times 0.6^2 \left[\frac{275^2 - N_{min}^2}{2} \right] \\
275^2 - N_{min}^2 &= 26597
\end{aligned}$$

$\therefore$ Speed after riveting, $N_{min} = \mathbf{221.4 \text{ rpm}}$

Rate of rivetting =

$$\begin{aligned}
&\frac{\text{Total energy supplied by motor}}{\text{Energy required for rivetting}} \\
&= \frac{P \times 3600}{E} = \frac{2750 \times 3600}{12000} = \mathbf{825 \text{ rivets/hour}}
\end{aligned}$$

Cross-sectional area of the rim, $A = bt$

But $b = 2t$

$$\therefore A = 2t^2$$

Mass of the flywheel, $m = \pi DA \times \rho$

$$150 = \pi \times 1.2 \times 2t^2 \times 7200 \Rightarrow t^2 = 2.763 \times 10^{-3}$$

Thickness, $t = 0.0525 \text{ m} = 5.25 \text{ cm}$

Breadth, $b = .105 \text{ m} = 10.5 \text{ cm}$



Chapter 4

Vibration

Put simply, vibration is nothing but a to and fro motion of a body about a mean position. Any motion that repeats itself after regular intervals of time is said to be **periodic**.

A simple pendulum executing small oscillations is a common example of an object



FIGURE 1. Examples of vibrating bodies

in periodic motion. Similarly if a cantilever beam with an end mass is given an initial disturbance as shown, it vibrates on its own in a periodic fashion.

Moreover, the simple pendulum and cantilever beam that we discussed are also examples of simple harmonic motion.

4.1

Simple Harmonic Motion (SHM)

The kind of periodic motion in which the acceleration of the object is always directly proportional to the displacement from the mean position is called **Simple Harmonic Motion** or **SHM**.

This can be easily visualized by considering a rotating vector OP of length X as given below.



FIGURE 2. SHM

Vector OP rotates anti-clockwise about O. After turning an angle θ , the projection of OP on the vertical axis YY' is given by $x = OP' = OP \sin \theta = X \sin \theta$.

Now if we plot the projections in the Y-axis and time (t) in the X-axis, we will get a sine curve. At any given instant of time (t), the vertical displacement x is given by $X \sin \theta$ where θ is the angle turned during time 't'. $\theta = \omega t$ where ω is the angular velocity of vector OP.

Let us look at the up and down motion of P' along YY'.

Displacement, $x = X \sin \omega t$

The first time derivative $\frac{dx}{dt}$ will be denoted by $\dot{x}$ and second time derivative $\frac{d^2x}{dt^2}$ will be denoted by $\ddot{x}$

Velocity, $\frac{dx}{dt}$, $\dot{x} = \omega X \cos \omega t$

Acceleration, $\frac{d^2x}{dt^2}$, $\ddot{x} = -\omega^2 X \sin \omega t = -\omega^2 x$

Since $-\omega^2$ is a constant here, $\ddot{x} \propto x$.

i.e. acceleration is directly proportional to displacement. Hence the to and fro motion of the vertical projection of vector OP is analogous to any object executing simple harmonic motion. The negative sign in $-\omega^2 x$ hints that the motion is directed towards the mean position.

Alternately, the standard equation for SHM is

$$\ddot{x} + \omega^2 x = 0$$

If we start with the projection on the horizontal axis, the equation $x = X \cos \omega t$ can be used and we will get the same result $\ddot{x} \propto x$.

4.2

Terms used in the study of vibration

Some of the key terms used in the study of vibration are defined below

Amplitude, X - It is the maximum displacement from the mean position. Refer fig. When θ becomes 90 degrees or $\frac{\pi}{2}$ radians, $OP' = OP = X$. This corresponds to the crest of the sine wave. It can be seen that at $\frac{3\pi}{2}$ radians there is the trough of the wave, and this too, represents the amplitude of the wave.

Cycle - The motion of the body from mean position to an extreme position, again to mean, then to the other extreme and back to mean is called a complete cycle. This is represented by a cycle from 0 radians to 2π radians in the sine wave.

Time Period, T - It is the time taken to complete one cycle. The vector OP completes one full revolution, i.e, 2π rad.

If the angular velocity of the vector OP is ω , the time period is given as

$$T = \frac{2\pi}{\omega}$$

Frequency, f - It is the number of cycles per unit time. It is the reciprocal of time period.

$$\text{Frequency, } f = \frac{1}{T} = \frac{\omega}{2\pi}$$

The unit of frequency is Hertz or cycles per second (*cps*). Sometimes we use ω as a measure of frequency since 2π is a constant. Unit of ω is rad/s.

Phase Difference - Consider two sine waves, one leading the other by an angle ϕ . The vector OP_1 leads the vector OP_2 by an angle of ϕ . This is the phase difference between the two waves. For the first wave, $x_2 = X \sin \omega t$ and $x_1 = X \sin (\omega t + \phi)$.



FIGURE 3. Phase difference

At the time 't' seconds, the first wave has reached the maximum, i.e., amplitude X while the second wave reached only $x' = X \sin \omega t$.

Wave 1 is always ahead of wave 2 and ϕ is called the phase difference between the two waves. Looking at the figure, one can see that when ϕ is 90 degrees, the first wave becomes a cosine wave.

Natural frequency, ω_n - If a system is given an initial disturbance and then left to vibrate on its own, it will do so in a particular frequency. This is called the natural frequency of the system denoted by ω_n or f_n . If the system has damping, then the corresponding natural frequency is called the damped natural frequency denoted by ω_d or f_d .

Damping - The resistance to vibration inherent in all vibratory systems is referred to as damping. The oscillation of simple pendulum eventually dies down due to damping offered by air resistance and friction in the hinges. Though damping is present in varying degrees in all real world systems, undamped vibrations are of academic interest.

Free and forced vibrations - Once a system is set to vibrate, is there is no application of external force, the system is free to vibrate in its own natural frequency. Such vibrations are called free vibrations.

Even after a system is set to vibrate, if there is a regular application of external force at definite intervals of time, the corresponding vibrations are called forced vibrations. At times, when the frequency of the force matches with the natural frequency of the system, the displacements become huge giving rise to a phenomenon called resonance which is discussed in detail in later sections.

4.3

Methods for solving vibration problems

There are two methods commonly used to solve vibration problems. They are

- (1) Energy Method
- (2) Newton's Method

4.4

Spring-Mass model

It is convenient to represent any vibrating system using a spring and mass model as shown. The spring of stiffness 'k' is fixed at one end and a mass 'm' suspended from the other.

An analysis of a simple spring-mass system is called for at this point.

Newton's Method

Figure a is a spring without any mass attached to it.

Figure b shows static equilibrium when the system is stationary with equal and



FIGURE 4. Spring-mass model

opposite forces canceling themselves.

$$k\delta = mg$$

Figure c shows the system vibrating about the mean position. If we draw the free-body diagram of the vibrating spring mass system when the mass is at an arbitrary 'x' distance from the mean,

$$-kx + k\delta - mg = m\ddot{x}$$

$$-kx = m\ddot{x}$$



The negative sign for spring force ($-kx$) is due to the fact that it is a restoring force and x is measured in the opposite direction.

From equation

$$m\ddot{x} + kx = 0$$

we get

$$\ddot{x} + \frac{k}{m}x = 0$$

The standard differential equation for SHM is

$$\ddot{x} + \omega^2 x = 0$$

Comparing the two,

$$\omega = \sqrt{\frac{k}{m}}$$

Energy Method

This method can be used only for conservative systems *i.e.* systems without damping. Since there is no dissipation of energy we can assume that the total energy of the system will be a constant at all times.

Potential energy of the spring, $PE = \frac{1}{2}kx^2$

Kinetic energy of the spring, $KE = \frac{1}{2}mv^2$

The entire stiffness of the system is assumed to be concentrated in the spring and the entire mass is assumed to be concentrated in the body m .

$$P.E. + K.E. = \text{constant}$$

$$\frac{1}{2}kx^2 + \frac{1}{2}mv^2 = C$$

Differentiating once,

$$kx\dot{x} + m\dot{x}\ddot{x} = 0$$

Canceling $\dot{x}$ throughout,

$$kx + m\ddot{x} = 0$$

Solution of the differential equation

$$m\ddot{x} + kx = 0$$

The solution of the above equation is

$$x = A\cos\omega_n t + B\sin\omega_n t \quad (4.1)$$

The values of A and B can be found out using initial conditions. At time $t=0$

$$x = x_0$$

$$\dot{x} = v_0$$

Substituting in equation 4.4,

$$x_0 = A\cos 0 + B\sin 0 = A$$

$$\dot{x} = -A\omega_n \sin\omega_n t + B\omega_n \cos\omega_n t$$

$$\text{At } t = 0, v_0 = \dot{x} = -A\omega_n \sin 0 + B\omega_n \cos 0 = B\omega_n$$

$$\text{So, } A = x_0 \text{ and } B = \frac{v_0}{\omega_n}$$

The solution, i.e. equation 4.4 becomes,

$$x = x_0 \cos \omega_n t + \frac{v_0}{\omega_n} \sin \omega_n t$$

Equivalent stiffness of springs

Let us take a look at the procedure for calculating the equivalent stiffness of springs connected in different configurations.

Springs in parallel Equivalent stiffness of springs connected in parallel can be found out as follows.

If k_1 and k_2 are to be replaced by a single spring of stiffness k_e then

The spring forces should add up and the displacements should be the same

$$k_e x = k_1 x + k_2 x$$

Canceling out the displacement x

$$k_e = k_1 + k_2$$

Springs in series If one is to replace k_1 and k_2 by an equivalent spring force k_e , then

The displacements should add up and the spring force should be the same

$$x_e = x_1 + x_2$$

$$\frac{F_s}{k_e} = \frac{F_s}{k_1} + \frac{F_s}{k_2}$$

Canceling out the spring force F_s

$$\frac{1}{k_e} = \frac{1}{k_1} + \frac{1}{k_2}$$

4.5

Solved Problems - Undamped free vibration

PROBLEM 4.1. Find the natural frequency of a simple pendulum having a length l and mass m shown in the figure below.

SOLUTION. This problem can be solved using either Newton's method or Energy method. Both are given below.

Newton's method

Torque = Moment of inertia $\times$ Angular acceleration

$$\tau = I\alpha$$

τ = Force $\times$ displacement between the axis of rotation and the line of action of the force = $mg \times l \sin \theta$

$$I = ml^2 \text{ (about } O \text{)}$$

So, $mg l \sin \theta = ml^2 \alpha = ml^2 \ddot{\theta}$. For small oscillations, $\sin \theta \approx \theta$. Therefore,

$$mg l \theta = ml^2 \ddot{\theta}$$

$$\text{i.e. } \ddot{\theta} + \frac{g}{l} \theta = 0$$

Compare the above equation with the standard SHM equation $\ddot{x} + \omega_n^2 x = 0$.

We have

$$\omega_n = \sqrt{\frac{g}{l}}$$

In Hertz,

$$f_n = \frac{1}{2\pi} \sqrt{\frac{g}{l}}$$

Energy Method

When the pendulum bob swings through an angle θ the height to which the pendulum bob is raised is given by $H = l - l \cos \theta$ as can be seen from the given figure.

Potential Energy, P.E. = $mgH = mg(l - l \cos \theta)$

Kinetic Energy, K.E. = $\frac{1}{2} I \omega^2 = \frac{1}{2} ml^2 \dot{\theta}^2$

$$K.E. + P.E. = \text{constant}$$

$$\frac{1}{2} ml^2 \dot{\theta}^2 + mg l (1 - \cos \theta) = C$$

Differentiating,

$$ml^2 \dot{\theta} \ddot{\theta} + mg l \sin \theta \dot{\theta} = 0$$

Putting $\sin\theta \approx \theta$ and canceling $ml\ddot{\theta}$ throughout,

$$l\ddot{\theta} + g\theta = 0$$

$$\ddot{\theta} + \frac{g}{l}\theta = 0$$

Comparing with the standard equation for SHM $\ddot{x} + \omega_n^2 x = 0$, again we have

$$\omega_n = \sqrt{\frac{g}{l}}$$

In Hertz,

$$f_n = \frac{1}{2\pi} \sqrt{\frac{g}{l}}$$

PROBLEM 4.2. Determine the governing equation as well as frequency of oscillation of the system shown in figure below. Assume pure rolling without slipping.

SOLUTION. **Energy Method**

Potential Energy, P.E. = $\frac{1}{2}kx^2$ (Potential Energy of the spring)

Kinetic Energy, K.E. = $\frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$ (Rotational and translational)

From figure, we have $x = r\theta$. Therefore, $\dot{x} = r\dot{\theta}$ and $\ddot{x} = r\ddot{\theta}$

$$K.E. + P.E. = \text{constant}$$

$$\frac{1}{2}m\dot{x}^2 + \frac{1}{2}I\dot{\theta}^2 + \frac{1}{2}kx^2 = \text{constant}$$

$$\frac{1}{2}m\dot{x}^2 + \frac{1}{2}\left(\frac{1}{2}mr^2\right)\left(\frac{\dot{x}}{r}\right)^2 + \frac{1}{2}kx^2 = \text{constant}$$

Differentiating,

$$m\dot{x}\ddot{x} + \frac{1}{2}m\dot{x}\ddot{x} + kx\dot{x} = 0$$

Cancel $\dot{x}$ throughout.

$$\frac{3}{2}m\ddot{x} + kx = 0$$

On rearranging, we get

$$\ddot{x} + \frac{2k}{3m}x = 0$$

as the governing equation. Comparing this with the standard equation of SHM $\ddot{x} + \omega_n^2 x = 0$, we get

$$\omega_n = \sqrt{\frac{2k}{3m}} \quad \text{and} \quad f_n = \frac{1}{2\pi} \sqrt{\frac{2k}{3m}}$$

Newton's Method

Identify all forces and draw the free body diagram.

Resultant of all forces = Mass $\times$ acceleration

$$\begin{aligned} F_s - f &= ma \\ k(-x) - f &= m\ddot{x} \\ -kx - f - m\ddot{x} &= 0 \end{aligned} \quad (4.2)$$

Taking moments about O, $f \times r = I\alpha$ where f is the frictional force. i.e.

$$\begin{aligned} f \times r &= \frac{1}{2}mr^2\ddot{\theta} = \frac{1}{2}mr^2\left(\frac{\ddot{x}}{r}\right) \\ f &= \frac{1}{2}m\ddot{x} \end{aligned} \quad (4.3)$$

Substitute (2) in (1)

$$\begin{aligned} kx + \frac{1}{2}m\ddot{x} + m\ddot{x} &= 0 \\ \frac{3}{2}m\ddot{x} + kx &= 0 \end{aligned}$$

On rearranging we get the governing equation

$$\ddot{x} + \left(\frac{2k}{3m}\right)x = 0$$

Comparing this with the standard equation of SHM $\ddot{x} + \omega_n^2x = 0$, we get

$$\omega_n = \sqrt{\frac{2k}{3m}} \quad \text{and} \quad f_n = \frac{1}{2\pi} \sqrt{\frac{2k}{3m}}$$

PROBLEM 4.3. *Find the natural frequency of oscillation of the system shown in figure.*

SOLUTION. *Proceed as in the previous example (Problem no.2).*

The expression for natural frequency is same as that for the system in the previous example.

PROBLEM 4.4. *Find out the natural frequency of the given system if $k_1 = k_2 = k_3 = k_4 = k_5 = k_6 = 100 \text{ N/m}$*

SOLUTION. *This can be simplified as*

and

where

$$\frac{1}{k_e} = \frac{1}{300} + \frac{1}{300} = \frac{1}{150}$$

Therefore $k_e = 150 \text{ N/m}$

From the analysis of a simple spring mass system, the natural frequency is given by

$$\omega = \sqrt{\frac{k_e}{m}} = \sqrt{\frac{150}{10000}} = 0.122 \text{ rad/s}$$

PROBLEM 4.5. What is the governing equation and natural frequency of oscillation of the liquid column in the U-tube shown below?

SOLUTION. Here the restoring force that causes the oscillation is the weight of the displaced portion of the liquid column of height $2x$.

If the density of the liquid is ρ and the cross-sectional area of the tube is a , then weight of the displaced portion of liquid column $= \rho a (2x) g = 2\rho a x g$.

Mass oscillating $=$ total mass of the liquid $= \rho a l$. So,

$$-2\rho a x g = \rho a l \ddot{x}$$

Rearranging, the governing equation is obtained as

$$\ddot{x} + \frac{2g}{l}x = 0$$

From this,

$$\omega_n = \sqrt{\frac{2g}{l}} \quad \text{and} \quad f_n = \frac{1}{2\pi} \sqrt{\frac{2g}{l}}$$

PROBLEM 4.6. Find the natural frequency of the system shown in the figure.

SOLUTION. From the geometry, we can have a relation connecting the displacement x of the spring and the angle θ .

$$\tan\theta = \frac{x}{d}$$

Restoring moment is given by

$$-mg(l\sin\theta) - kx(d) = -mgl\sin\theta - kd^2\tan\theta$$

But restoring moment is equal to

(Moment of inertia about point O) $\times$ (Angular acceleration)

$$-mgl\sin\theta - kd^2\tan\theta = ml^2\ddot{\theta}$$

When θ is very small, $\sin\theta \rightarrow \theta$, $\cos\theta \rightarrow 1$, $\tan\theta \rightarrow \theta$.

$$ml^2\ddot{\theta} + (mgl + kd^2)\theta = 0$$

$$\ddot{\theta} + \frac{mgl + kd^2}{ml^2}\theta = 0$$

Comparing this with the standard equation of SHM $\ddot{x} + \omega_n^2 x = 0$, we get

$$\omega_n = \sqrt{\frac{mgl + kd^2}{ml^2}}$$

PROBLEM 4.7. Derive the governing equation and the expression for natural frequency of the given system.

SOLUTION. A relation between the displacement x of the mass m and the angle θ turned by the pulley can be deduced.

Assume that the mass m comes x distance downwards. This x is divided between the two sides of the string and consequently the center of the pulley O comes $\frac{x}{2}$ distance down.

Then the spring will be extended by $\frac{x}{2}$

$$R\theta = \frac{x}{2} ; \theta = \frac{x}{2R}$$

Kinetic energy of the system, $K.E$

$$= \frac{1}{2}m\dot{x}^2 + \frac{1}{2}\left(\frac{MR^2}{2}\right)\dot{\theta}^2 + \frac{1}{2}M\left(\frac{\dot{x}}{2}\right)^2$$

where $\frac{MR^2}{2}$ is the moment of inertia of the pulley.

Potential energy, $P.E = \frac{1}{2}k\left(\frac{x}{2}\right)^2$

Total energy is conserved i.e. $K.E. + P.E. = \text{constant}$.

Differentiating with respect to time,

$\frac{d}{dt}(KE + PE) = 0$ i.e.

$$m\dot{x}\ddot{x} + \left(\frac{MR^2}{2}\right)\dot{\theta}\ddot{\theta} + k\frac{x}{4}\dot{x} + \frac{M\dot{x}\ddot{x}}{4} = 0$$

But $\theta = \frac{x}{2R}$. So,

$$m\dot{x}\ddot{x} + \left(\frac{MR^2}{2}\right)\frac{\dot{x}}{2R}\frac{\ddot{x}}{2R} + \frac{kx}{4}\dot{x} + \frac{M\dot{x}\ddot{x}}{4} = 0$$

Canceling $\dot{x}$ throughout the equation,

$$\left[m + \frac{M}{8} + \frac{M}{4}\right]\ddot{x} + \left(\frac{k}{4}\right)x = 0$$

The governing equation becomes

$$\ddot{x} + \frac{2k}{8m + 3M}x = 0$$

From this, the natural frequency

$$\omega_n = \sqrt{\frac{2k}{8m + 3M}}$$

PROBLEM 4.8. Find out the natural frequency of compound pendulum.

SOLUTION. In simple pendulum assumption, we neglect the mass of the rod. In compound pendulum assumption, we consider the mass of the rod also.

Let the mass of the rod be M and the mass of the bob be m .

Restoring torque $= -mgl \sin \theta - Mg \frac{l}{2} \sin \theta$

But according to Newton's Law, restoring torque $= I_0 \ddot{\theta}$

where

$$I_0 = [MI \text{ of the rod about } O + MI \text{ of the bob about } O] = \frac{1}{3}Ml^2 + ml^2$$

Thus,

$$\begin{aligned} -mgl\theta - Mg\frac{l}{2}\theta &= \left(\frac{1}{3}Ml^2 + ml^2\right)\ddot{\theta} \quad \sin \theta \approx \theta \\ \ddot{\theta} + \frac{mg + \frac{Mg}{2}}{\frac{1}{3}Ml + ml}\theta &= 0 \end{aligned}$$

On rearranging,

$$\ddot{\theta} + \frac{1 + \frac{Mg}{2m}}{1 + \frac{M}{3m}} \left(\frac{g}{l}\right) \theta = 0$$

Comparing this with the standard equation of SHM $\ddot{x} + \omega_n^2 x = 0$, we get

$$\omega_n = \sqrt{\frac{g}{l} \left[\frac{1 + \frac{M}{2m}}{1 + \frac{M}{3m}} \right]}$$

PROBLEM 4.9. Find the equivalent spring constant for the given system. Assume $k_1 = k_2 = k_3 = k_4 = k_5 = 1 \text{ N/cm}$

SOLUTION. Springs k_1 and k_2 in parallel can be replaced by a spring of stiffness, $k_{12} = k_1 + k_2 = 2 \text{ N/cm}$.

Then, k_{12} and k_3 in series can be replaced by a spring of stiffness

$$\frac{k_{12}k_3}{k_{12} + k_3} = \frac{2 \times 1}{2 + 1} = \frac{2}{3} \text{ N/cm}$$

The arrangement will be as shown.

The two springs in parallel can be replaced by a spring of stiffness

$$\frac{2}{3} + 1 = \frac{5}{3} \text{ N/cm}$$

Now the arrangement will be like

Since the deflection for these springs will be equal, the two springs are in parallel. So, the equivalent spring stiffness is given by

$$\frac{5}{3} + 1 = \frac{8}{3} N/cm$$

PROBLEM 4.10. Determine the governing equation and natural frequency of vibration for the given system. Take the polar moment of inertia of the compound pulley as J .

SOLUTION. $x = 2r\theta$ and $x_2 = r\theta$

Kinetic energy, $K.E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}J\dot{\theta}^2$

Potential energy, $P.E = \frac{1}{2}kx_2^2$

$K.E + P.E = \text{Constant}$

$$\frac{1}{2}m(2r\dot{\theta})^2 + \frac{1}{2}J\dot{\theta}^2 + \frac{1}{2}kx_2^2 = \text{Constant}$$

Differentiating with respect to time and canceling $\dot{\theta}$ throughout

$$4mr^2\ddot{\theta} + J\ddot{\theta} + kr^2\theta = 0$$

Rearranging, we get

$$\ddot{\theta} + \frac{kr^2}{4mr^2 + J}\theta = 0$$

which is comparable to the standard equation for SHM. Hence, the natural frequency

$$\omega_n = \sqrt{\frac{kr^2}{4mr^2 + J}}$$

PROBLEM 4.11. What is the governing equation and natural frequency of vibration of mass m ? Assume that the tension in the string is constant and is T .

SOLUTION. Let the mass be displaced slightly i.e. x from the horizontal configuration. See fig. ??.

The restoring force is given by $T \cos \alpha + T \cos \beta$

For small angles,

$$\cos \alpha = \frac{x}{a} \quad \text{and} \quad \cos \beta = \frac{x}{l-a}$$

From Newton's Law,

$$-(T \cos \alpha + T \cos \beta) = m\ddot{x}$$

i.e., $m\ddot{x} + T[\cos \alpha + \cos \beta] = 0$

$$m\ddot{x} + T\left[\frac{x}{a} + \frac{x}{l-a}\right] = 0$$

On rearranging, the governing equation is obtained as

$$\ddot{x} + \frac{T}{m} \left\{ \frac{l}{a(l-a)} \right\} x = 0$$

From the governing equation, the natural frequency

$$\omega_n = \sqrt{\frac{Tl}{ma(l-a)}}$$

PROBLEM 4.12. A cylinder rolls on a curved surface of radius R without slipping as shown in fig. ?? . Find the natural frequency of oscillation.

SOLUTION. From the geometry, $R\theta = r\phi$

So,

$$\phi = \frac{R}{r}\theta$$

Angle turned along the curved surface $= \phi - \theta$

Angular velocity,

$$\omega = \dot{\phi} - \dot{\theta} = \left(\frac{R}{r} - 1 \right) \dot{\theta}$$

Linear velocity of centre O ,

$$v = (R - r) \dot{\theta}$$

Total kinetic energy

$$\begin{aligned} &= \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 \\ &= \frac{1}{2}m \left\{ (R - r) \dot{\theta} \right\}^2 + \frac{1}{2} \left(\frac{mr^2}{2} \right) \left\{ \left(\frac{R}{r} - 1 \right) \dot{\theta} \right\}^2 \end{aligned}$$

Potential energy =

$$mgh = mg[(R - r)(1 - \cos \theta)]$$

Refer figure for geometry $K.E. + P.E = \text{constant}$

$$\frac{d}{dt}(K.E + P.E) = 0$$

$$\frac{1}{2}m(R - r)^2 2\dot{\theta}\ddot{\theta} + \frac{1}{2} \left(\frac{mr^2}{2} \right) \left(\frac{R - r}{r} \right)^2 2\dot{\theta}\ddot{\theta} + mg(R - r)\theta\dot{\theta} = 0$$

Canceling common terms and combining,

$$\frac{3}{2}(R - r)\ddot{\theta} + g\theta = 0$$

$$\ddot{\theta} + \frac{2g}{3(R - r)}\theta = 0$$

The natural frequency

$$\omega_n = \sqrt{\frac{2g}{3(R-r)}}$$

PROBLEM 4.13. Determine the natural frequency of the system in fig ?? assuming the beam OB massless.

SOLUTION. Assuming unit weight at P (i.e. $W = 1N$) and taking moments about Q ,

$$k_2 x_2 \times (a + b) = 1 \times a$$

Similarly, taking moments about R

$$k_1 x_1 \times (a + b) = 1 \times b$$

Therefore, deflection of spring k_1 and k_2 are

$$x_1 = \frac{b}{(a+b)k_1}, \quad x_2 = \frac{a}{(a+b)k_2}$$

The configuration of the arrangement as it is deflected slightly is given by figure ??
From the law of similar of triangles,

$$\frac{y}{a} = \frac{d}{(a+b)}$$

$$y = \frac{ad}{(a+b)} = \frac{a}{(a+b)}(x_2 - x_1)$$

Deflection of point P , $= x_1 + y =$

$$\begin{aligned} & \frac{b}{(a+b)k_1} + \frac{a}{(a+b)} \left[\frac{a}{(a+b)k_2} - \frac{b}{(a+b)k_1} \right] \\ &= \frac{1}{(a+b)^2} \left[\frac{a^2}{k_2} + \frac{b^2}{k_1} \right] \end{aligned}$$

Force at P has been assumed as 1 Newton. So, the equivalent spring stiffness at point P is given by

$$\frac{1}{\frac{1}{(a+b)^2} \left[\frac{a^2}{k_2} + \frac{b^2}{k_1} \right]} = \frac{(a+b)^2}{\left[\frac{a^2}{k_2} + \frac{b^2}{k_1} \right]}$$

PROBLEM 4.14. Find the equivalent spring constant for the given system.

SOLUTION. Consider the portion OA when the point A is displaced by x_2 as shown. From the geometry, $X_2 = l_2 \theta$

The moment exerted by spring k_2 about O

$$= k_2 x_2 \times l_2 = k_2 l_2^2 \theta$$

An equivalent spring should exert the same moment. If this spring (i.e. k_2) is replaced with a spring of stiffness k hanging from B (l_1 from O), then the moment would $kl_1^2\theta$.

Equating,

$$\begin{aligned} k_2 l_2^2 \theta &= k l_1^2 \theta \\ k &= k_2 \left(\frac{l_2}{l_1} \right)^2 \end{aligned} \quad (4.4)$$

Now the equivalent system shown here can be solved as two springs in series.

$$\begin{aligned} \frac{1}{k_e} &= \frac{1}{k_1} + \frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2 \left(\frac{l_2}{l_1} \right)^2} \rightarrow (\text{from eqn 4.4}) \\ &= \frac{k_1 + k_2 \left(\frac{l_2}{l_1} \right)^2}{k_1 k_2 \left(\frac{l_2}{l_1} \right)^2} \end{aligned}$$

Natural frequency

$$\omega_n = \sqrt{\frac{k_e}{m}} = \sqrt{\frac{k_1 k_2 \left(\frac{l_2}{l_1} \right)^2}{m \left[k_1 + k_2 \left(\frac{l_2}{l_1} \right)^2 \right]}}$$

4.6

Damped Free Vibration

Damping can be defined as any influence on a vibrating system which reduces or prevents the vibrations. Due to damping, the energy of oscillation gets dissipated as heat, sound etc. All real systems have damping. Vibration which occurs without an external exciting force in the presence of damping is called *Damped free vibration*.

Here, a dashpot is provided in parallel with the spring as shown.

Damping force,

$$F_d = c\dot{x}$$

Spring force,

$$F_s = kx$$

According to Newton's Law,

Restoring force = Mass $\times$ Acceleration

$$-(F_s + F_d) = m\ddot{x}$$

The sign is negative because the force is opposite to the direction of x .

$$-kx - c\dot{x} = m\ddot{x}$$

or

$$m\ddot{x} + c\dot{x} + kx = 0 \quad (4.5)$$

Equation 4.5 is the governing differential equation for this system. This can be rewritten as

$$\ddot{x} + \frac{c}{m}\dot{x} + \frac{k}{m}x = 0 \quad (4.6)$$

Assuming a solution $x = Ae^{st}$
we have

$$\dot{x} = sAe^{st}\ddot{x} = s^2Ae^{st}$$